34721/34722 Linear Control Design 1
Spring Semester

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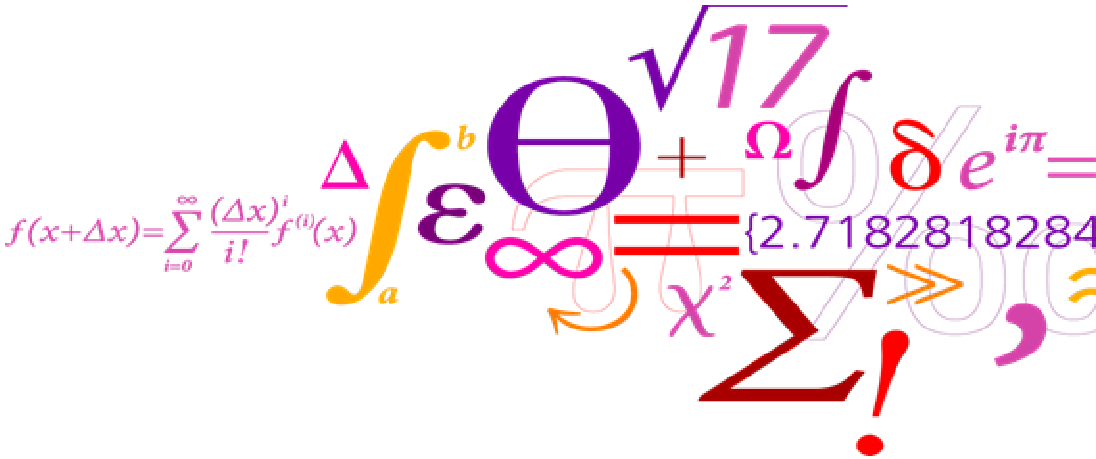
Lecture 7/13

Schedule (15:00-17:00)

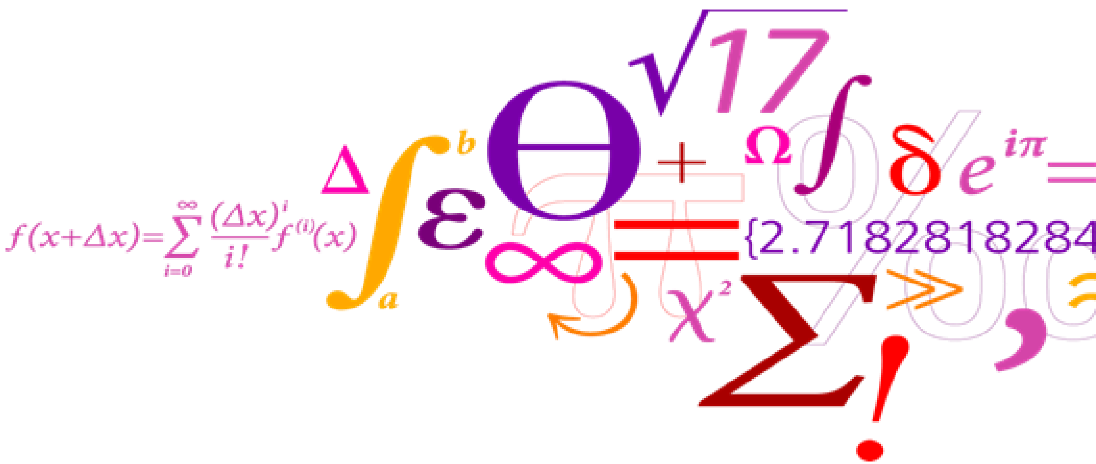
- **Basic concepts (recap)**
- **Stability margins (recap)**
- **Nyquist plot and stability criterion**



What you will learn today

After this lecture you will be able to

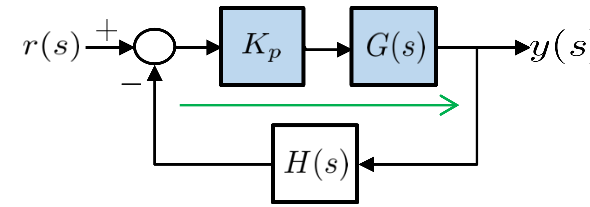
- Interpret the Nyquist plot of a linear system.
- Determine the stability margins of a system based on its Nyquist plot.
- Apply the Nyquist stability criterion to evaluate the closed-loop stability.



Some basics (recap)

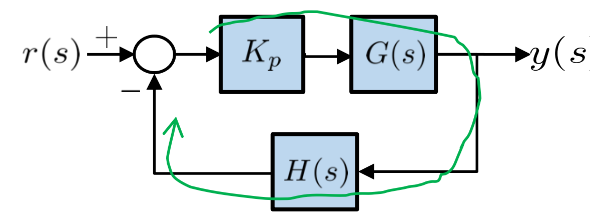
- **Forward branch** : From the summation to the output

$$G_{fb}(s) = K_p G(s)$$



- **(Open) Loop transfer function**: Make a look from the summation and all the way back to it and multiply whatever transfer function you find.

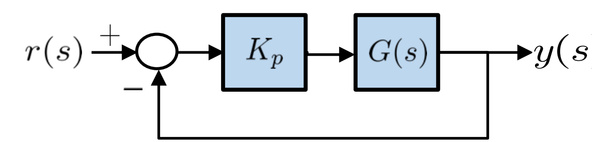
$$G_{ol}(s) = K_p G(s) H(s)$$



- **Closed-loop transfer function**: (Forward branch)/(1 + loop transfer function).

$$G_{cl}(s) = \frac{K_p G(s)}{1 + K_p G(s) H(s)}$$

- **Unit feedback**: Forward branch = Open loop transfer function

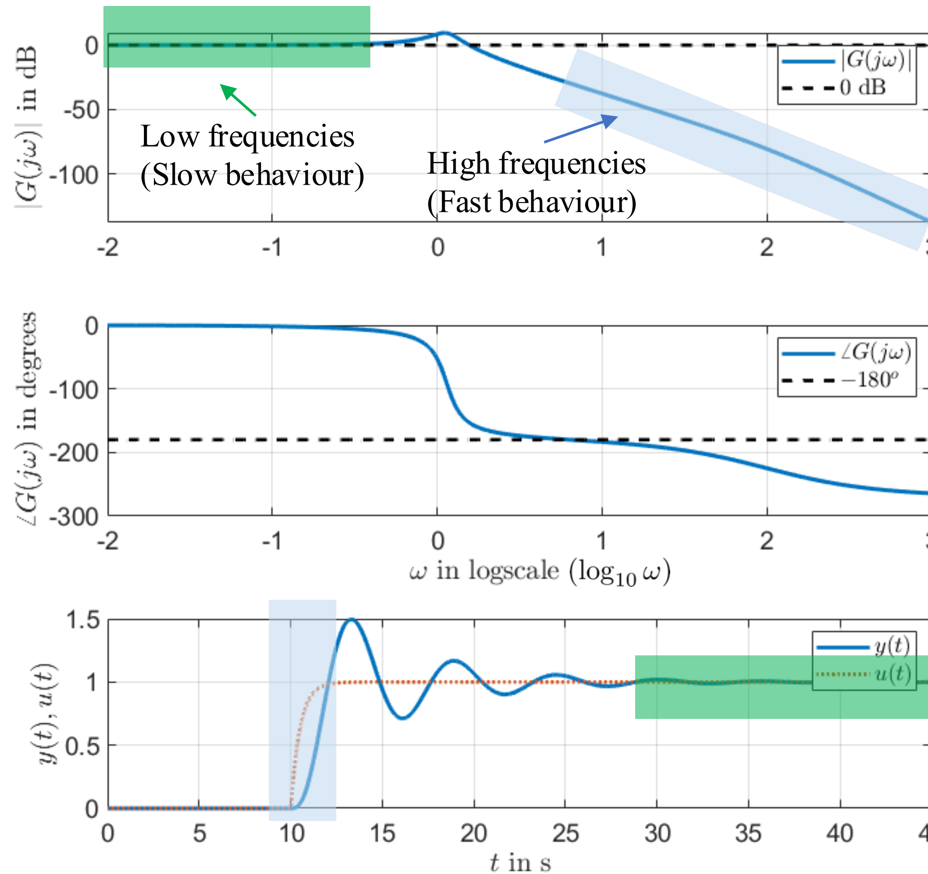


$$G_{cl}(s) = \frac{G_{ol}(s)}{1 + G_{ol}(s)}$$

- If the feedback signal comes with a "+", then the closed-loop transfer function is given by: $G_{cl}(s) = \frac{G_{ol}(s)}{1 - G_{ol}(s)}$

Some basics (recap)

- When we say “<u>low frequencies</u>” or steady-state, we mean the behaviour of the system when things have settled or when the input changes are slow, e.g. a slow sinusoidal
  → We look at the **left of the Bode plot**
- When we say “<u>high frequencies</u>”, we mean the behaviour of the system when the input changes are fast, e.g. a very early after a step change
  → We look at the **right of the Bode plot**
- The magnitude Bode plot shows how the output scales in size with respect to the input at every frequency.
- The phase Bode plot shows how much the output lags with respect to the input at every frequency. A -180^o lag means reversal of direction



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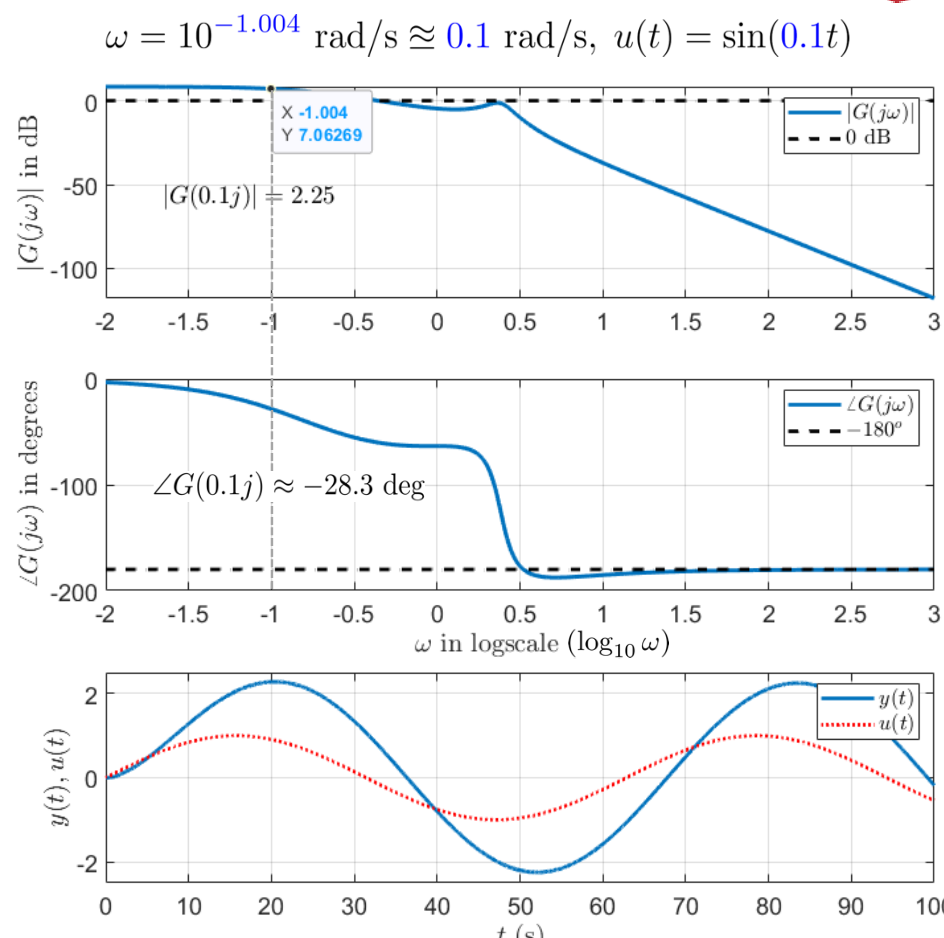

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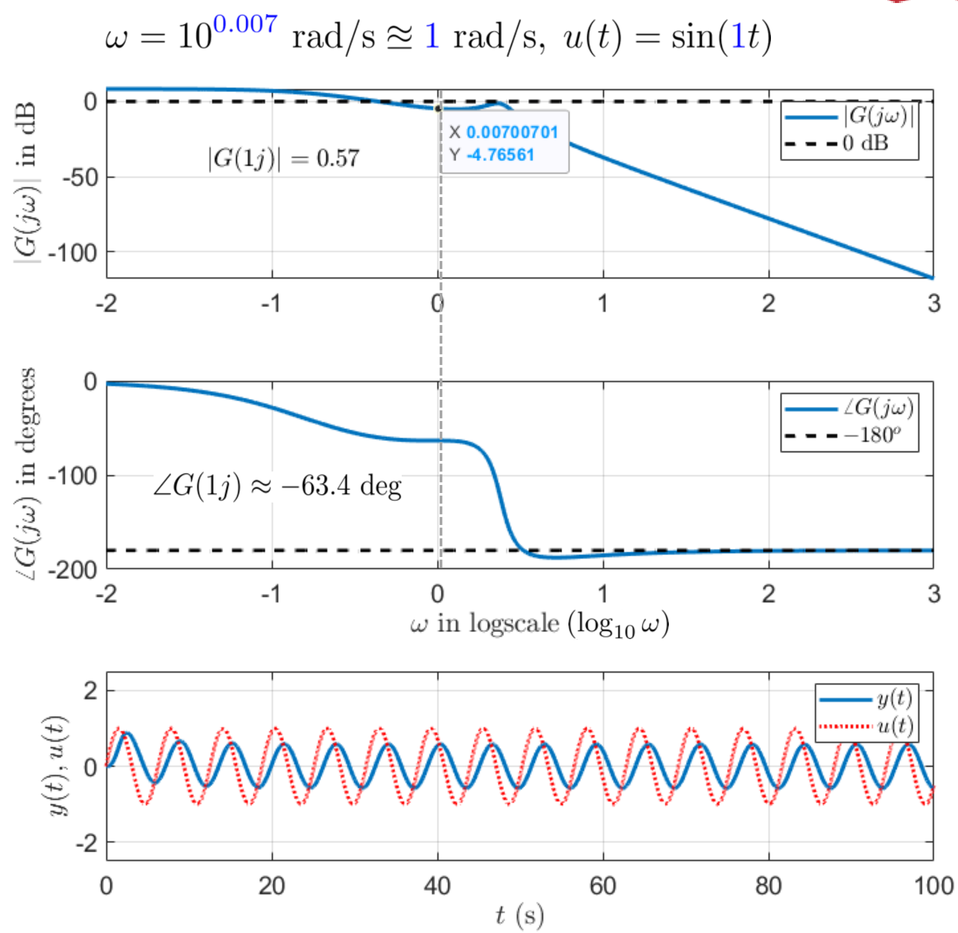

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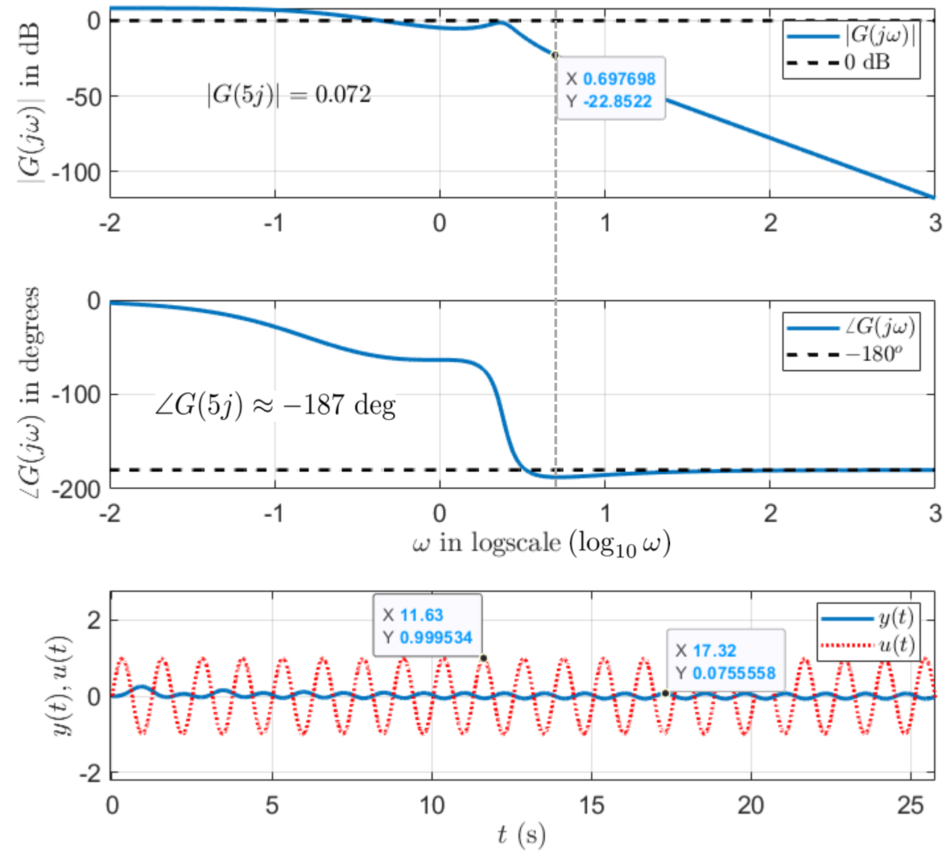
$\omega = 10^{0.697}$ rad/s ≈ 5 rad/s, $u(t) = \sin(5t)$



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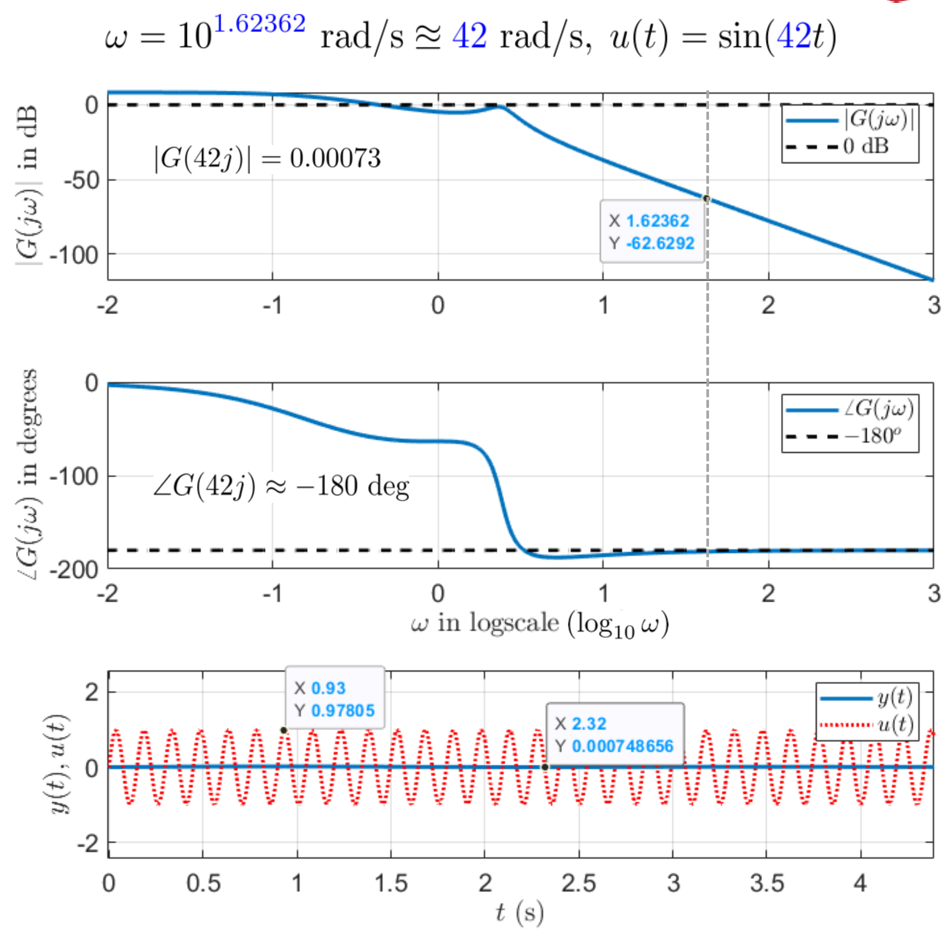
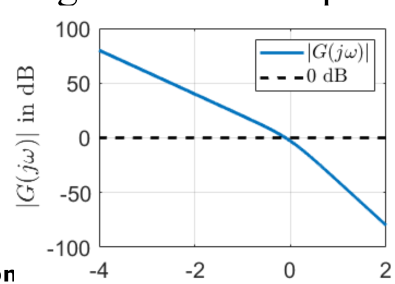


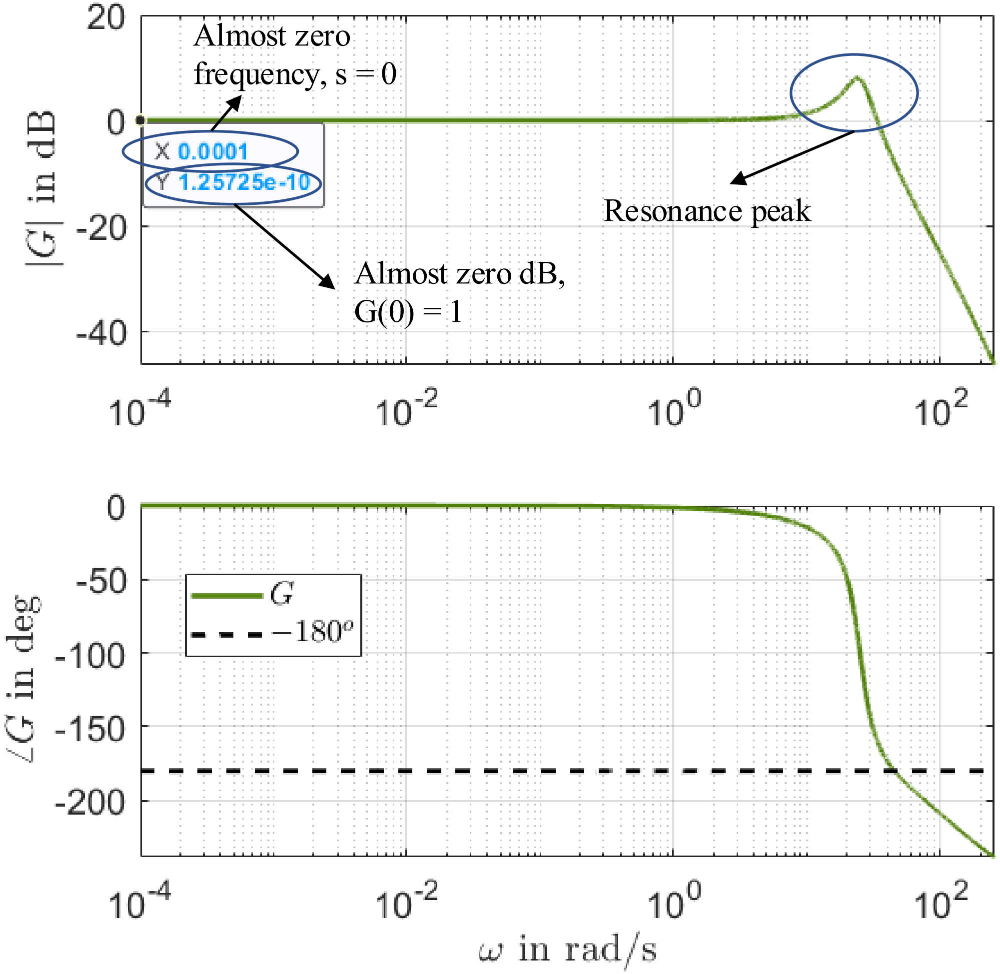
Some basics (recap)

Flat magnitude line at very low (almost zero) frequencies means constant steady-state gain.

- This is something we want to see in a closed-loop system Bode plot
- In **closed-loop systems** the longer the flat line is (the more it extends to higher frequencies), the better → the output follows faster references
- No pole at 0 (pure integrator) in the transfer function. If there were, the gain would be infinite in steady state, i.e. ever increasing magnitude curve as we go to lower frequencies

$$G_{cl}(s) = \frac{1}{s(1+s)}$$



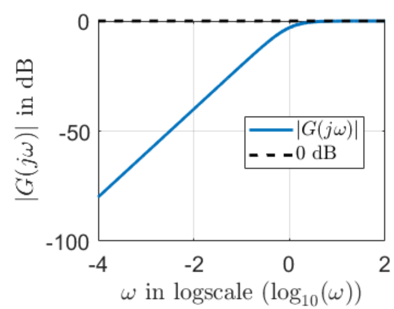


Some basics (recap)

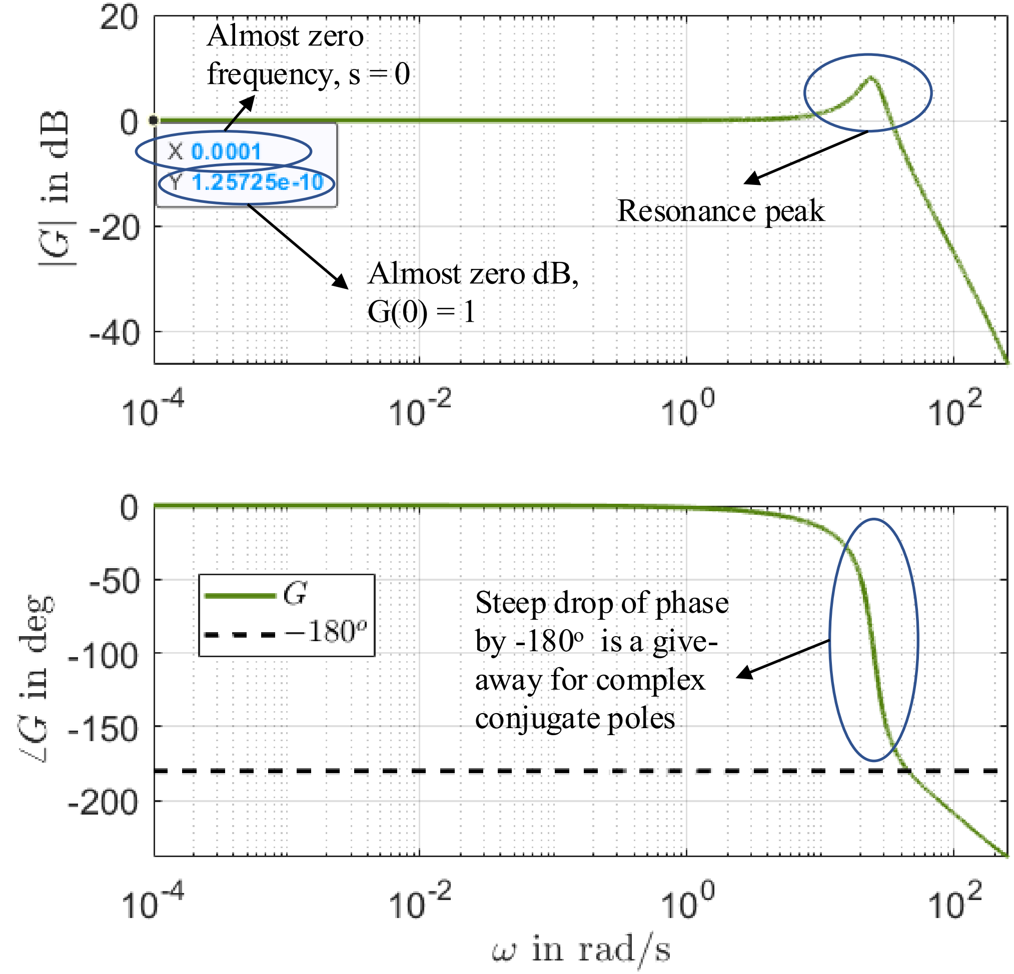
Flat magnitude line at very low (almost zero) frequencies means constant steady-state gain.

- No zero at 0 (pure differentiator) in the transfer function. If there were, the gain would be 0 in steady state, i.e. ever decreasing magnitude curve as we go to lower frequencies

$$G_{cl}(s) = \frac{s}{1+s}$$

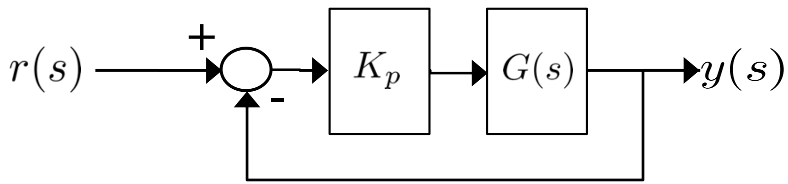


- **Resonance peak** implies complex poles and oscillatory response (in closed-loop Bode). We have <u>at least two</u> complex conjugates poles



Stability margins (recap)

Bode plot of <u>open-loop</u> transfer function can tell a lot about closed-loop stability



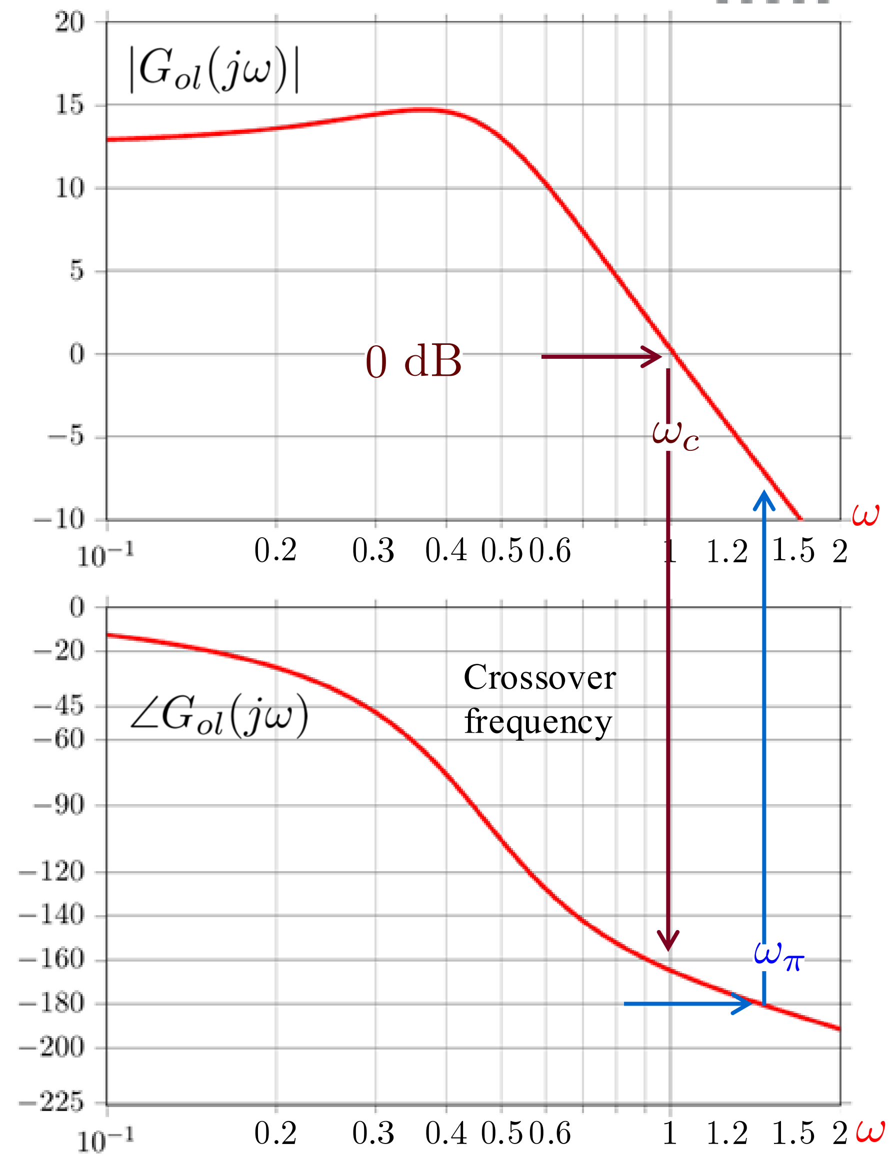
$$G_{ol}(s) = K_pG(s)$$

phase

$$G_{cl}(s) = \frac{K_pG(s)}{1+K_pG(s)} = \frac{G_{ol}(s)}{1+G_{ol}(s)}$$

- **Crossover frequency** (ω_c): Frequency at which the magnitude of G_{ol} crosses 0 dB gain.

- **π-frequency** (ω_π): Frequency at which the phase of G_{ol} crosses -180 degrees.



Stability margins (recap)

- Increasing/decreasing the open-loop gain by a factor of K ($K_p \rightarrow K \cdot K_p$) shifts the amplitude curve up/down by $20\log_{10}(K)$ units (dB).

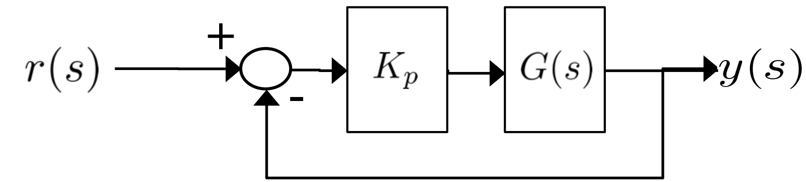
- ω_c is shifted to the right (increased gain) or to the left (decreased gain)

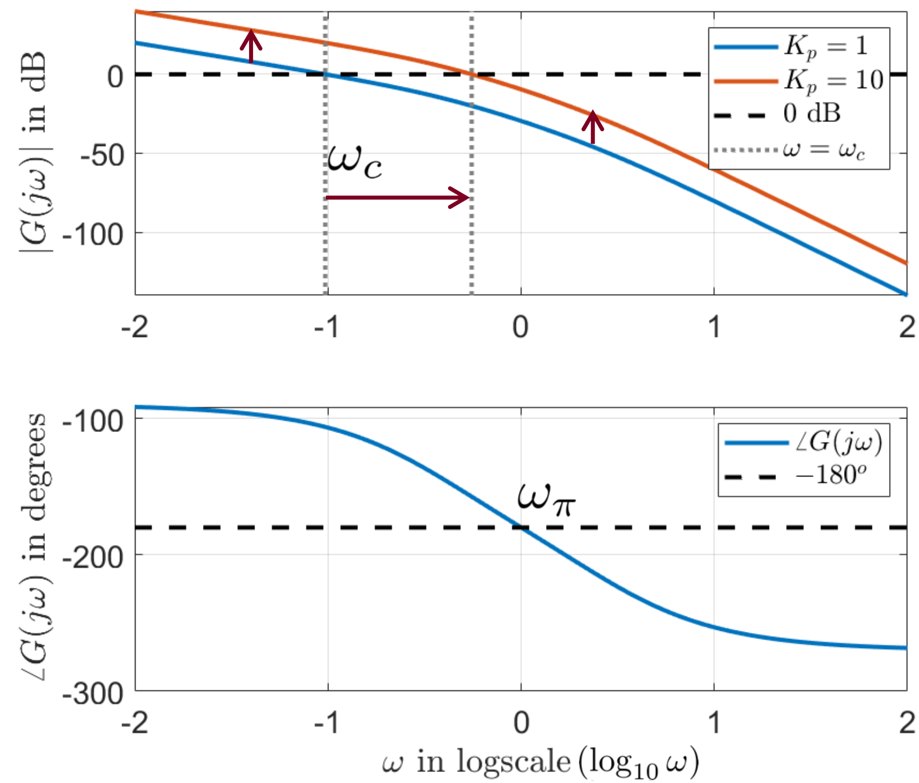
- The phase curve is not affected!

In **closed-loop** we have instability when

$$1 + G_{ol}(s) = 0 \Leftrightarrow G_{ol}(s) = -1 = 1 \cdot e^{-j\pi}$$

- We want $\omega_c < \omega_\pi$
- We cannot arbitrarily increase the open-loop gain





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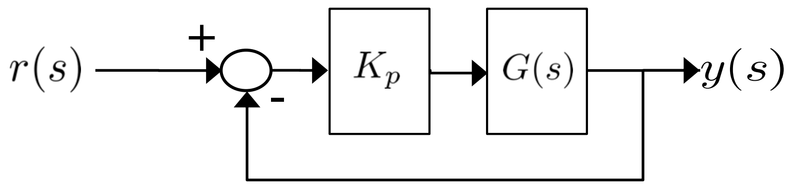
Bode plot of <u>open-loop</u> transfer function can tell a lot about closed-loop stability

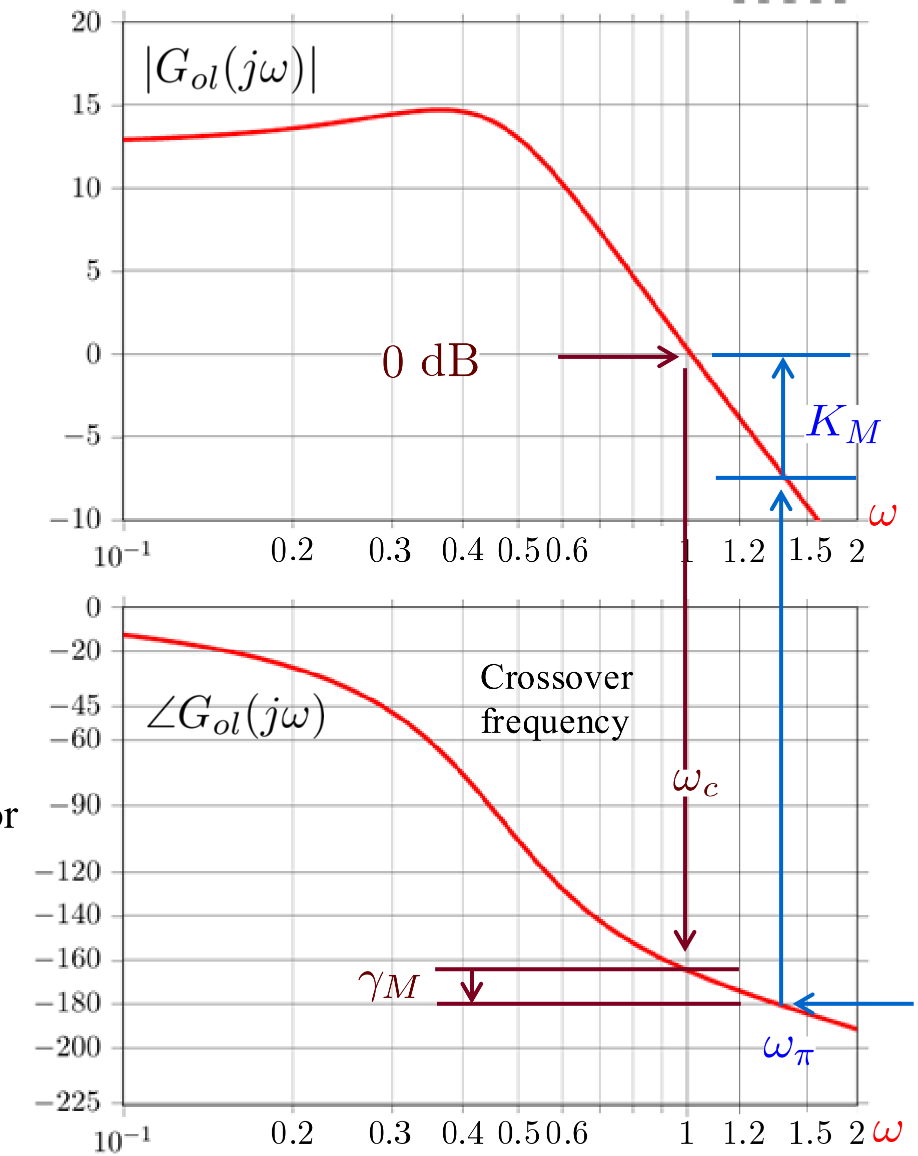


$$G_{ol}(s) = K_p G(s)$$

$$G_{cl}(s) = \frac{K_p G(s)}{1 + K_p G(s)} = \frac{G_{ol}(s)}{1 + G_{ol}(s)}$$

- **Gain margin** K_M : By what factor we need to multiply (or add in dB to) the open-loop gain until $\omega_c = \omega_\pi$.

- **Phase margin** γ_M : Difference in phase at ω_c and -180 degrees.



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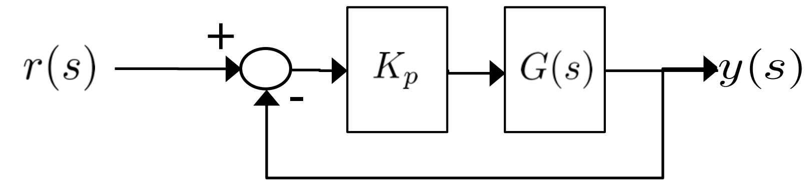

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<u>Example</u>

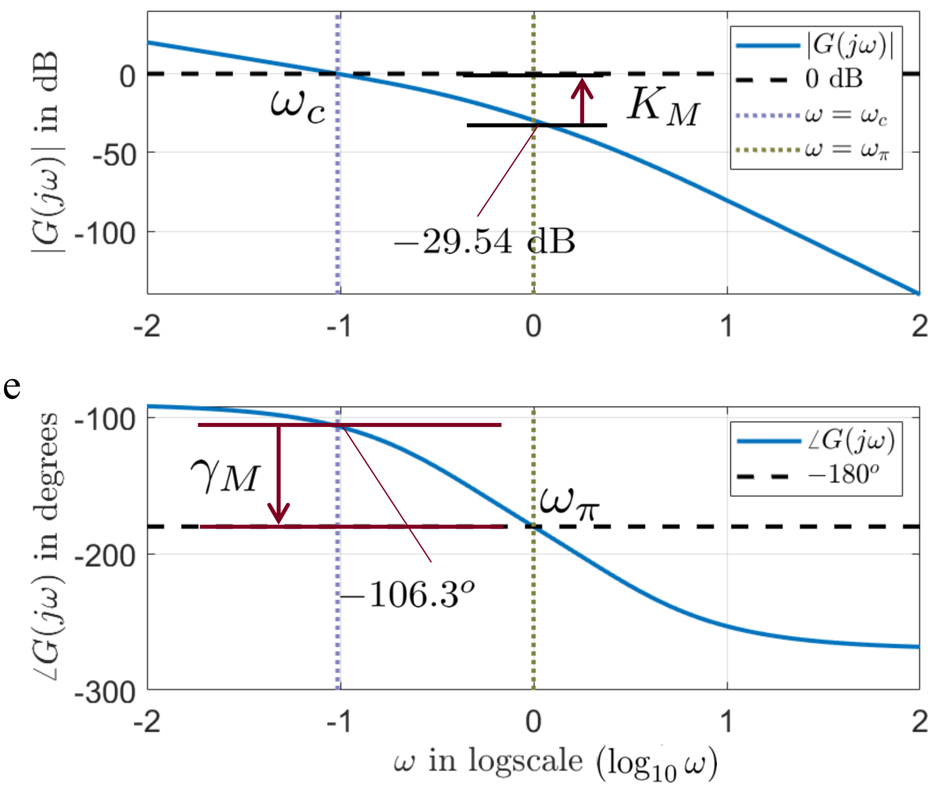
GM_PM.m

$$G(s) = \frac{0.1}{s(s^2+3s+1)}, \; K_p = 1, \; \omega_c \approx 0.1 \text{ rad/s}, \; \omega_\pi = 1 \text{ rad/s}$$

- At $\omega = \omega_\pi$, we have -29.54 dB magnitude → we can increase the K_p gain by 29.54 dB (or by a factor of $10^{\frac{29.54}{20}} \approx 30$) before we hit the stability limits ($\omega_c = \omega_\pi$)

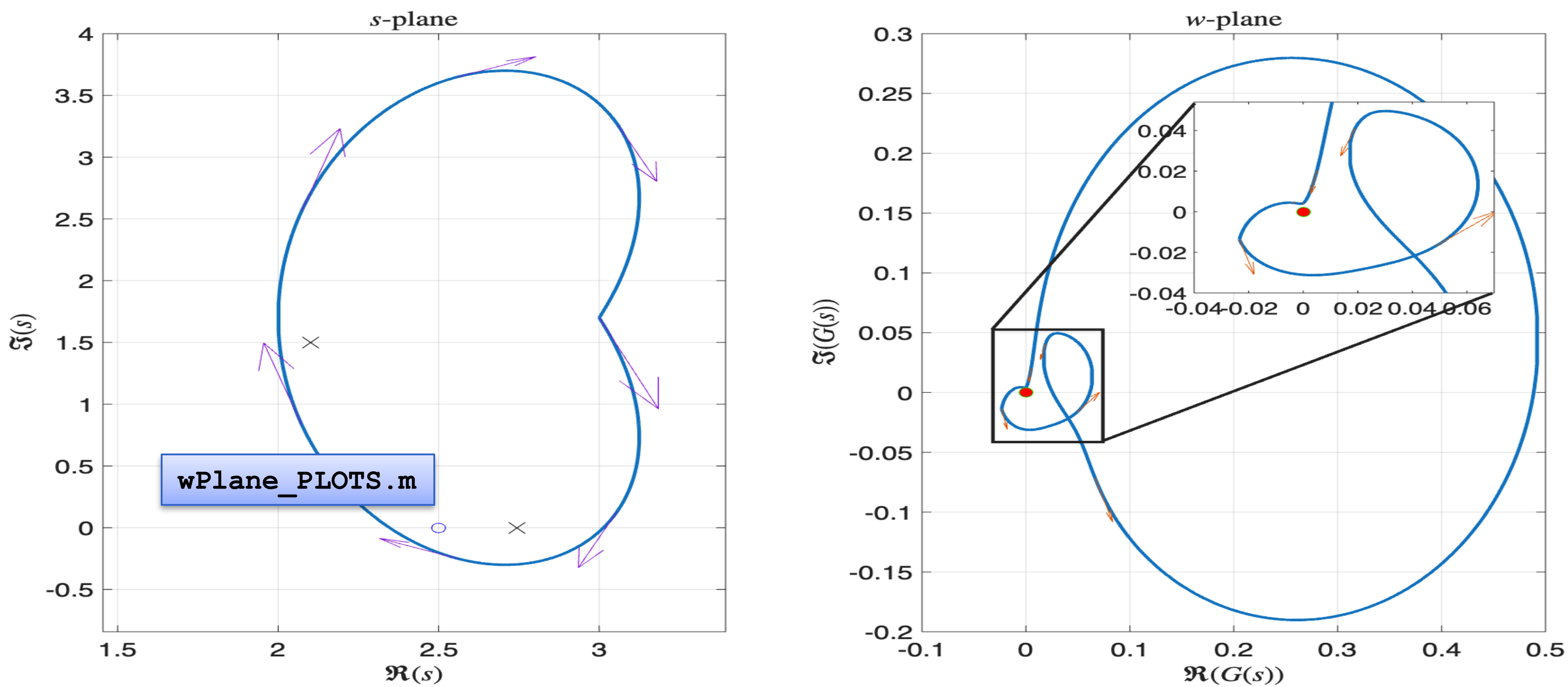
$$K_M^{dB} = 0 - (-29.54) = 29.54 \text{ dB or } K_M = 30$$

$$\gamma_M = -106.3^o - (-180^o) = 73.7^o$$



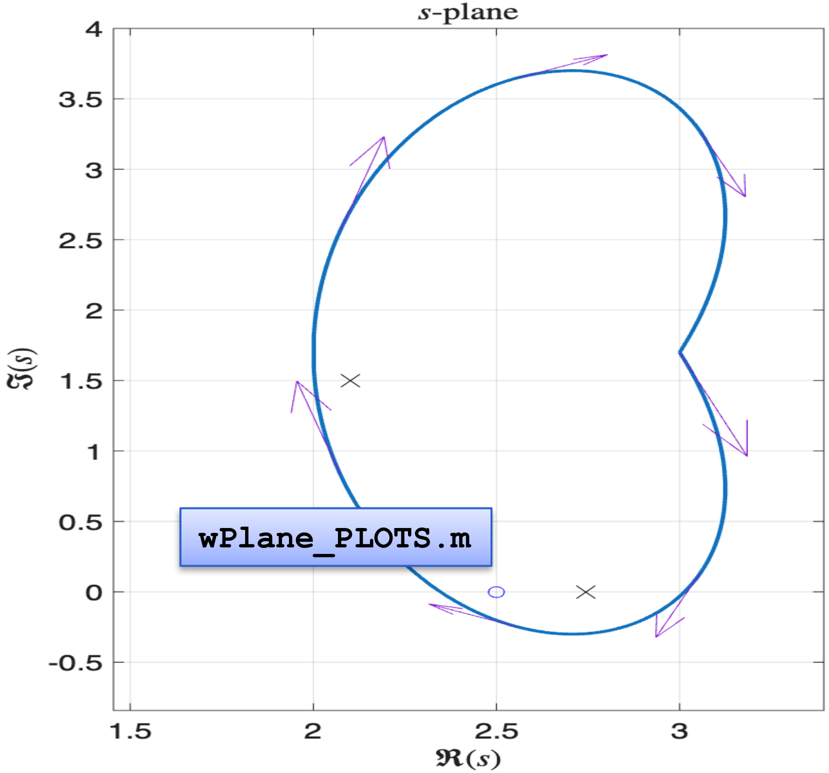
Nyquist plot

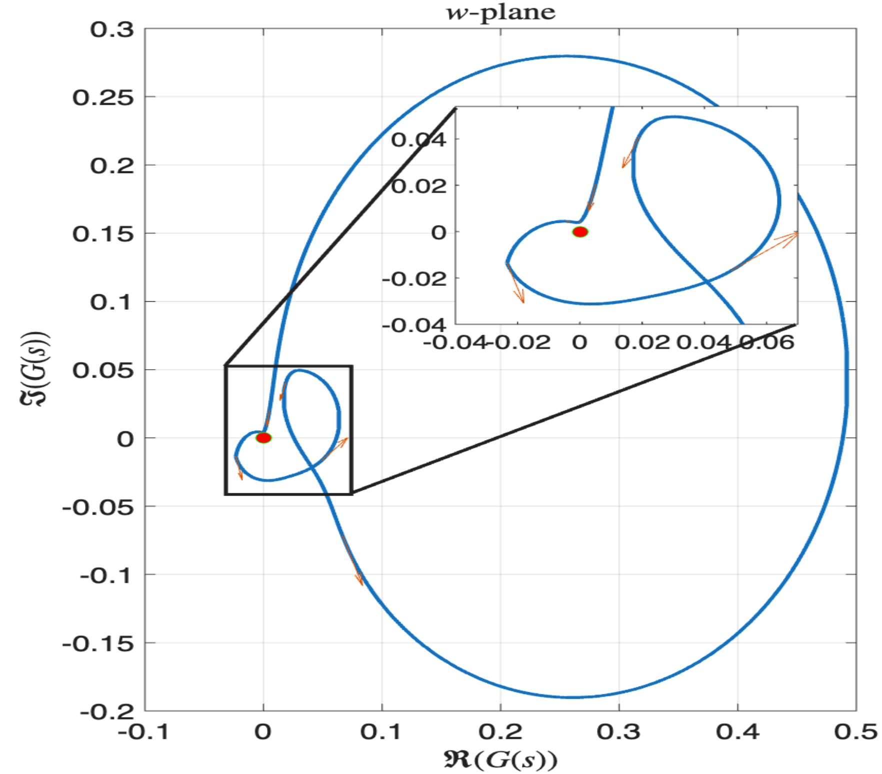
- Special type of **open-loop** plot that shows stability margins for closed-loop.
- Maps a closed curve $\mathbf{C_1}$ of points in *s*-plane to a closed curve $\mathbf{C_2}$ of points in the w-plane ($s \rightarrow G_{ol}(s)$).



Nyquist plot

- Select values for s on $\mathbf{C_1}$ and evaluate $G_{ol}(s)$ for these values → this will give $\mathbf{C_2}$
- For each zero of $G_{ol}(s)$ inside $\mathbf{C_1}$, the curve $\mathbf{C_2}$ will try to encircle the origin clockwise (CW)
- For each pole of $G_{ol}(s)$ inside $\mathbf{C_1}$, the curve $\mathbf{C_2}$ will try to encircle the origin counter-clockwise (CCW)





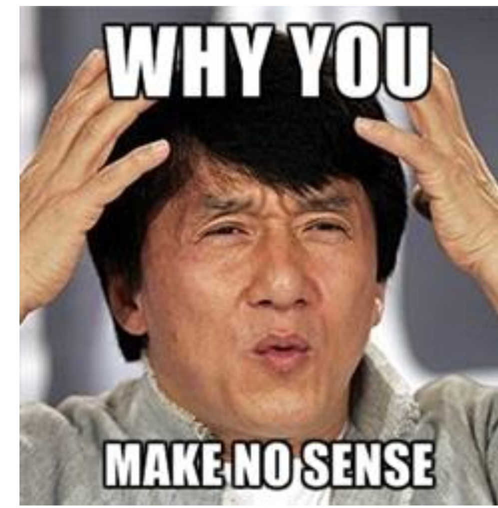
Nyquist plot – Determine closed-loop stability

If $\mathbf{C_1}$ includes the entire RHP, we can check whether ***$G_{ol}(s)$* encircles the origin**

- **"*$G_{ol}(s)$* encircles the origin"** → at least one value of s in the RHP that makes $G_{ol}(s)$ zero.

$$G_{cl}(s) = \frac{K_p G(s)}{1 + K_p G(s)} = \frac{G_{ol}(s)}{1 + G_{ol}(s)}$$

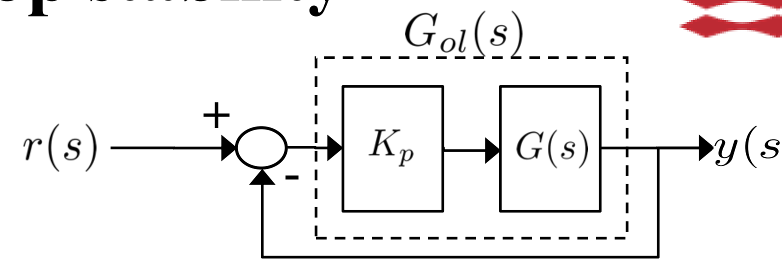
- The closed-loop is unstable when $1 + G_{ol}(s) = 0$
- What we need is a Nyquist plot for $1 + G_{ol}(s)$
- We can just make a Nyquist plot for $G_{ol}(s)$ and check for encirclements of -1.

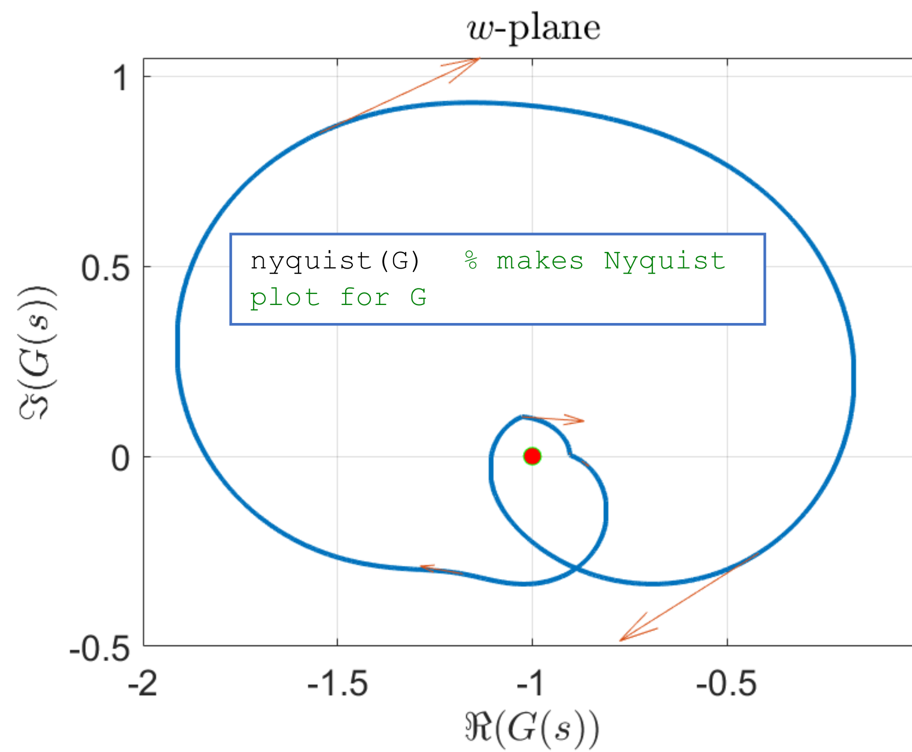


Nyquist plot – Determine closed-loop stability

1. Make Nyquist plot for $G_{ol}(s)$.
2. Check number of CW encirclements of -1, let it be N.
3. Check number of unstable poles of $G_{ol}(s)$, let it be P.
4. The total number of RHP zeros of $1 + G_{ol}(s)$ (unstable poles of $G_{cl}(s)$) Z is given by

$$Z = N + P$$



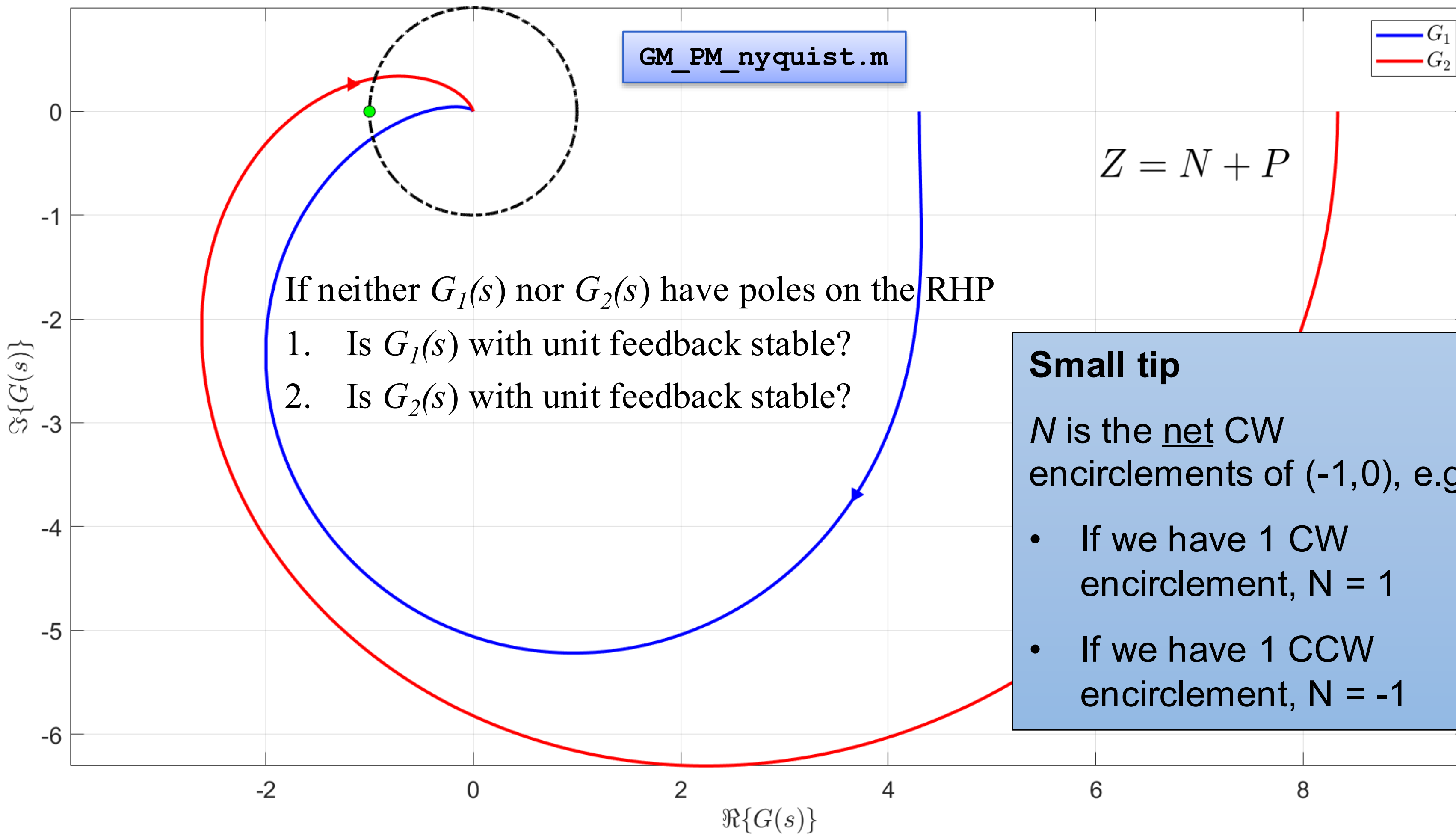


Mini quiz

If $G_{ol}(s)$ has no poles in the RHP

- Is $G_{cl}(s)$ stable?
- How many poles does $G_{cl}(s)$ have in the RHP?

Nyquist plot – Determine closed-loop stability



If neither $G_1(s)$ nor $G_2(s)$ have poles on the RHP

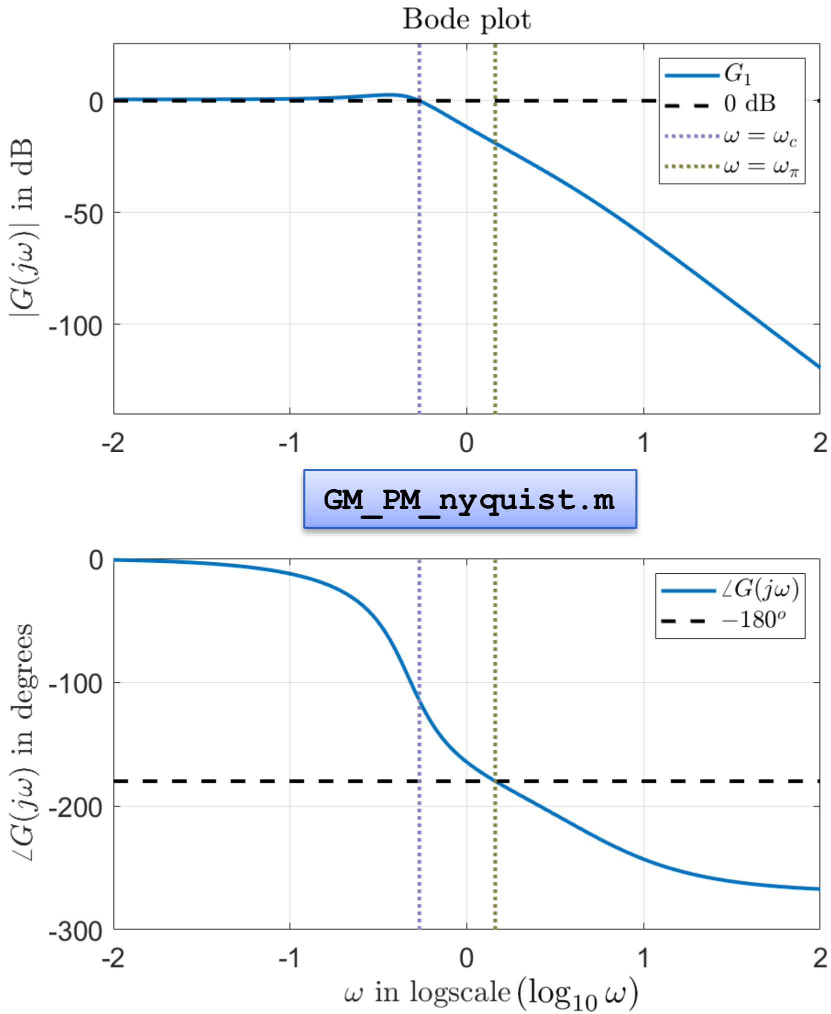
1. Is $G_1(s)$ with unit feedback stable?
2. Is $G_2(s)$ with unit feedback stable?

Small tip

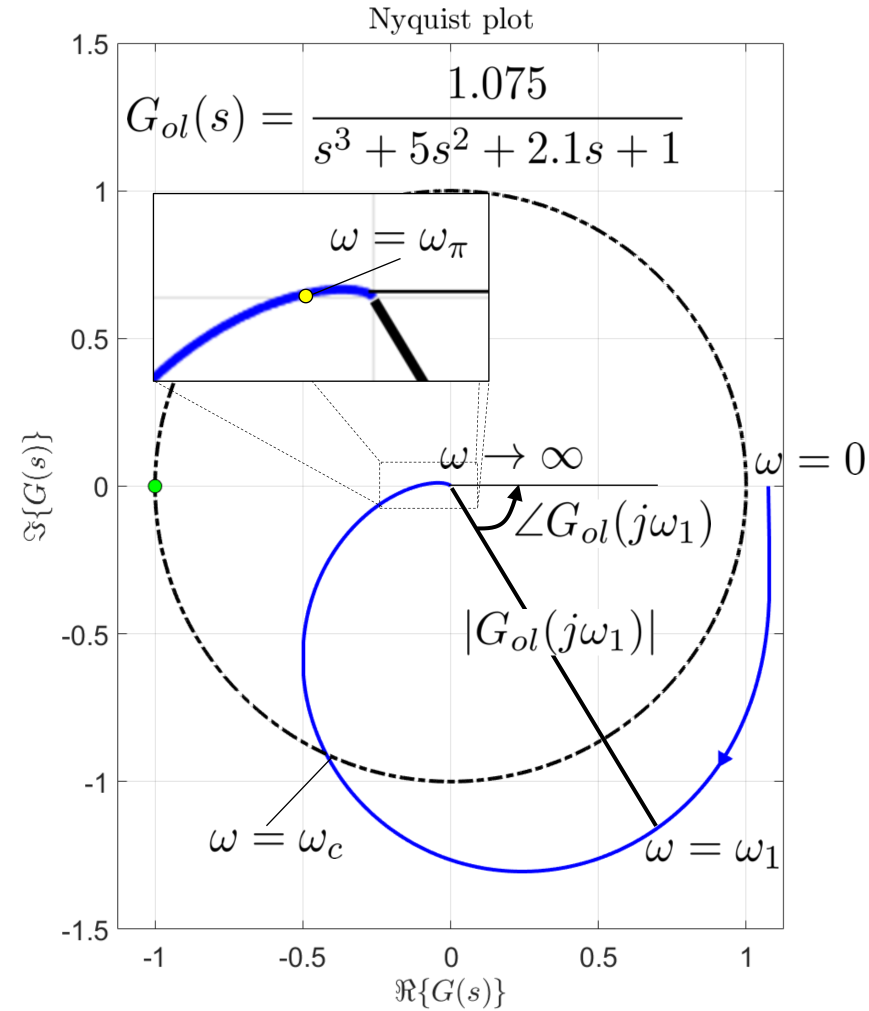
N is the <u>net</u> CW encirclements of (-1,0), e.g.

- If we have 1 CW encirclement, N = 1
- If we have 1 CCW encirclement, N = -1

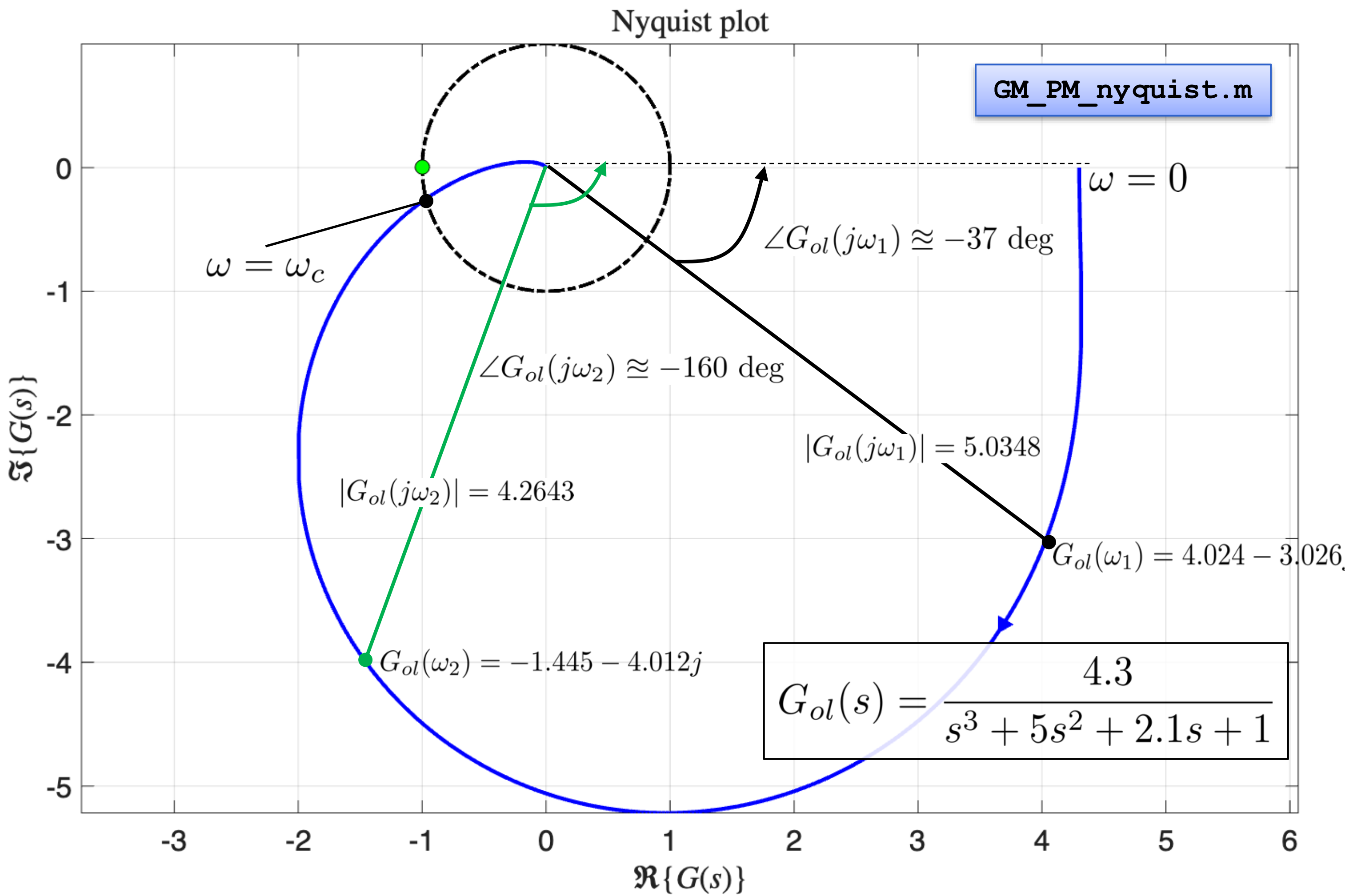
Nyquist plot – Determine closed-loop stability



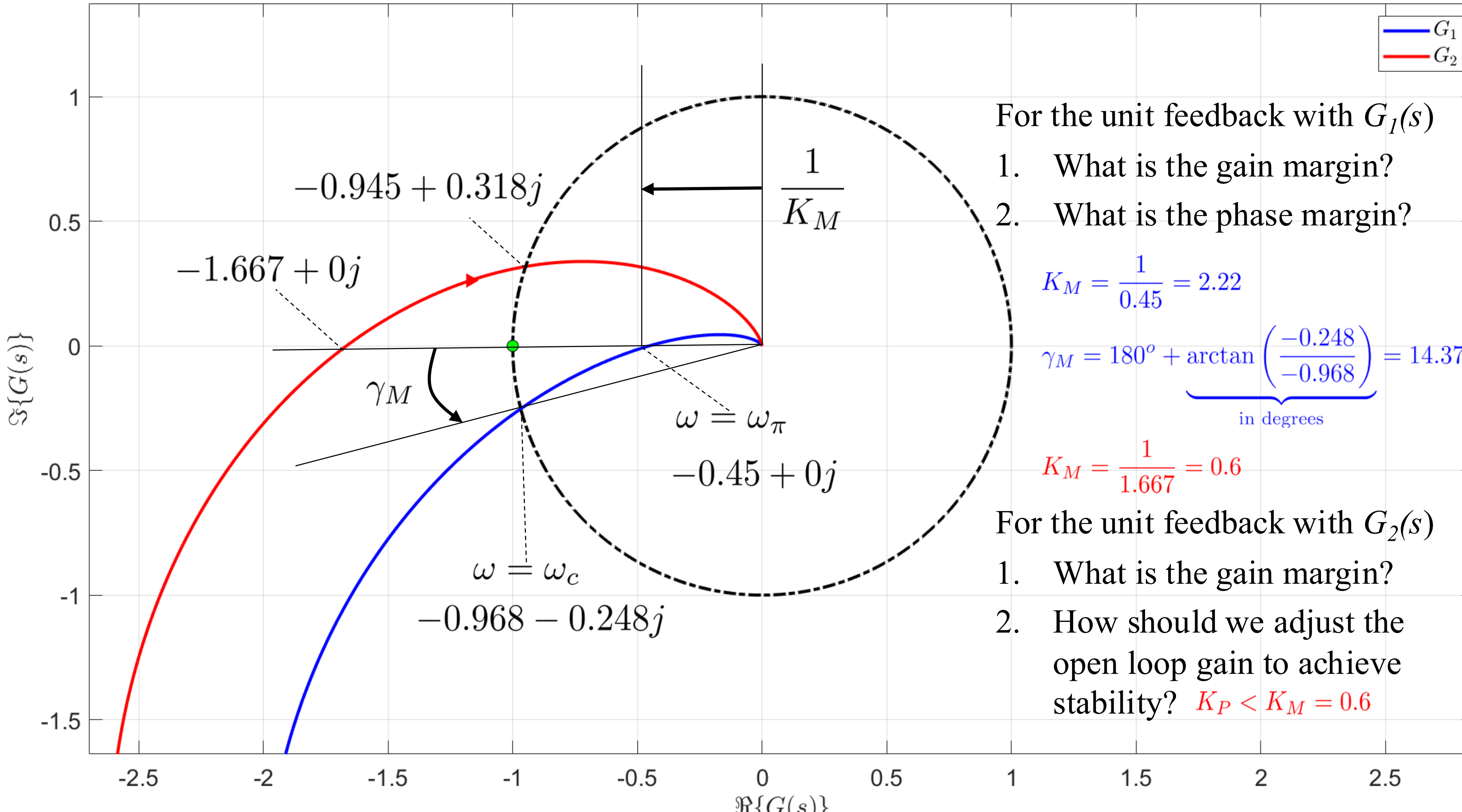
GM_PM_nyquist.m



Nyquist plot – Determine closed-loop stability



Nyquist plot – Determine closed-loop stability



For the unit feedback with $G_1(s)$

1. What is the gain margin?
2. What is the phase margin?

$$K_M = \frac{1}{0.45} = 2.22$$

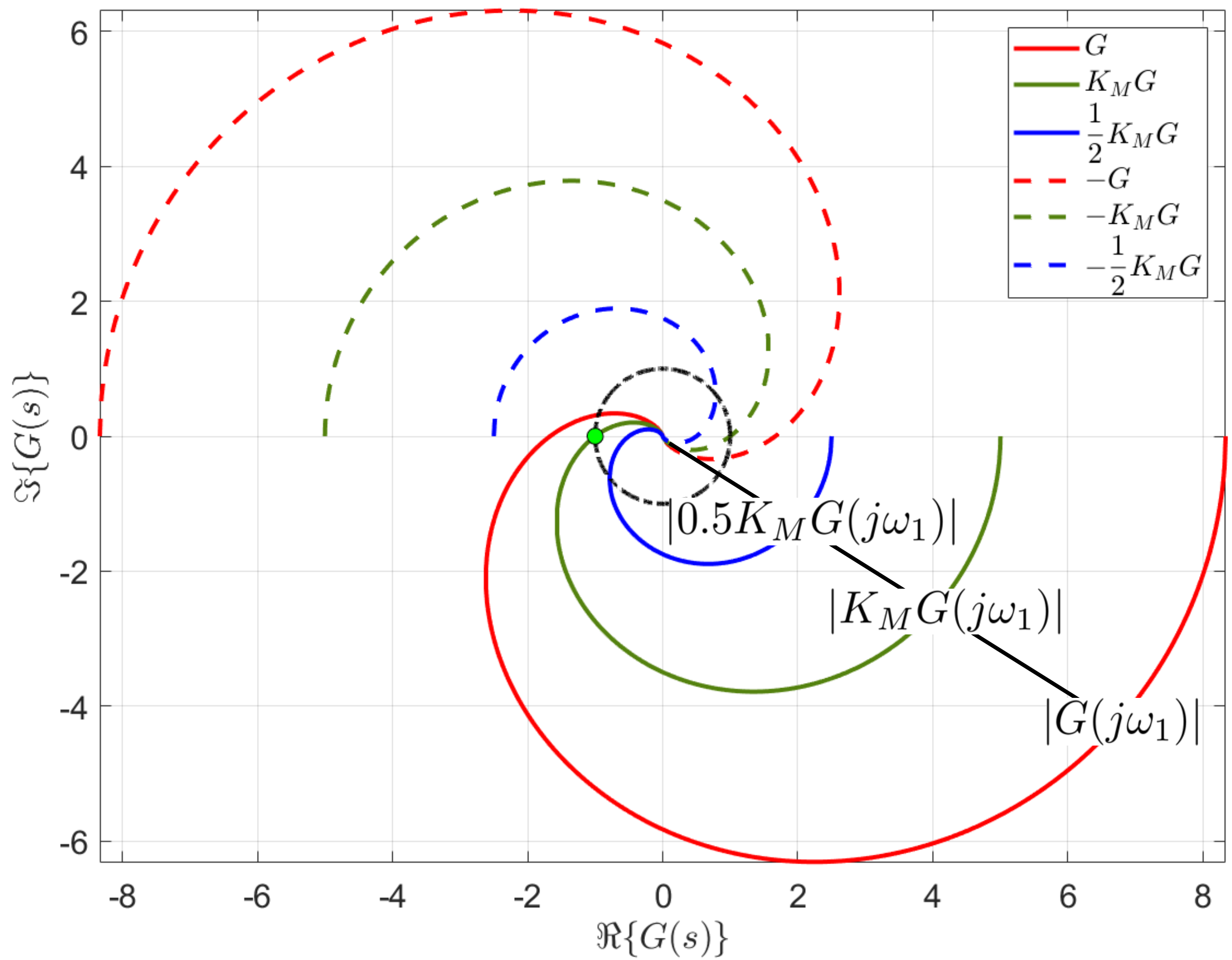
$$\gamma_M = 180^o + \underbrace{\arctan\left(\frac{-0.248}{-0.968}\right)}_{\text{in degrees}} = 14.37^o$$

$$K_M = \frac{1}{1.667} = 0.6$$

For the unit feedback with $G_2(s)$

1. What is the gain margin?
2. How should we adjust the open loop gain to achieve stability? $K_P < K_M = 0.6$

Nyquist plot manipulation



1. Multiplying with negative K_P gain flips the plot with respect to the origin.

2. Multiplying with smaller (larger) K_P gain shrinks (magnifies) the plot.

`Nyquist_example_gain_margin.m`

Today you learned

- What the Nyquist plot of a linear system is and how to produce it.
- How to determine the amplitude, phase and stability margins of a system based on its Nyquist plot.
- How evaluate the closed-loop stability by applying the Nyquist stability criterion.

Next lecture 8

PI-Lead control design

- Design a P controller
- Design a PI controller
- Design a PI-Lead controller